

Parameters and choices for AddiVortes

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Choice/parameter	Notation	Default value/Prior distribution
Metric to use	d	Euclidean
Number of tessellations	m	200
Variance of noise	σ	$\propto \frac{\nu\lambda^1}{\chi_\nu^2}$
Noise degrees of freedom hyperparameter	ν	3 – 10(6)
Noise alignment hyperparameter	q	0.7 – 0.99(0.85)
Number of covariates in T_j	d_j	$d_j - 1 \sim \text{Bin}(p - 1, \frac{\omega}{p})$
Number of centers in T_j	b_j	$b_j - 1 \sim \text{Po}(\lambda_c)$
Covariates included in T_j	—	Uniformly at random
Coordinates of centers of T_j	—	$\mathcal{N}(0, \sigma_c^2)^2$
Number of covariates hyperparameter	ω	
Number of centers hyperparameter	λ_c	
Center coordinates hyperparameter	σ_c	
Mean response of cell i under tessellation j	μ_{ij}	$\mathcal{N}(\mu_\mu, \sigma_\mu^2)$
Mean response center hyperparameter	μ_μ	0^3
Mean response spread hyperparameter	σ_μ	$\frac{0.5}{k\sqrt{m}}^4$
Mean response alignment hyperparameter	k	3
Probability of adding/removing a center	—	0.2
Probability of adding/removing a covariate	—	0.2
Probability of swapping a covariate	—	0.1
Probability of changing the position of a center	—	0.1
Number of MCMC iterations	—	1200
Number of MCMC burn-ins	—	200

¹Two plausible choices for $\hat{\sigma}$ are considered. The first is the “naive” option, wherein $\hat{\sigma}$ corresponds to the sample standard deviation of the transformed training response values. The second is the “linear model” choice, where $\hat{\sigma}$ is based on the residual standard deviation from a least-squares linear regression of scaled Y on the scaled X variables. Subsequently, a value for ν within the range of 3–10 is selected to shape the prior appropriately. Additionally, a suitable λ value is determined such that the qth quantile of the prior on σ aligns with $\hat{\sigma}$, that is $P(\sigma < \hat{\sigma}) = q$. A suitable range for q is values between 0.7 and 0.99.

²Drawn independently for each dimension

³Given the prior centered at 0, representing our zero-centered transformed data

⁴These hyper-parameters are set such that $m\mu_\mu - k\sqrt{m}\sigma_\mu = -0.5$ and $m\mu_\mu + k\sqrt{m}\sigma_\mu = 0.5$, where k is a preselected hyperparameter determining the prior probability of $E[Y \mid \mathbf{x}, \mathbf{y}_{\text{train}}, \mathbf{X}_{\text{train}}]$